Perturbative and Non-Perturbative Wings of the Zeta Building: Kreimer–Connes at $\zeta(s > 0)$ and Vacuum Cosmology at $\zeta(s < 0)$, Unified by the Functional Equation

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Abstract

The Kreimer–Connes Hopf algebra framework established that perturbative QFT renormalization produces motivic periods — specifically, multiple zeta values $\zeta(3), \zeta(5), \zeta(7), \ldots$ — as genuine physical content of loop integrals, not mere regularization artifacts. We observe that this "reification of ζ at positive integers" has a natural counterpart: the vacuum zero-point energy, regularized via ζ at *negative* integers ($\zeta(-1), \zeta(-3), \zeta(-5), \ldots$, equivalently Bernoulli numbers), also carries physical content. The Riemann functional equation $\zeta(s) \leftrightarrow \zeta(1-s)$ connects these two sides. We propose that the *full* physical content of the ζ function comprises both wings: the **perturbative wing** (Kreimer–Connes, ζ at positive integers, loop integrals) and the **non-perturbative wing** (vacuum cosmology, ζ at negative integers, boundary-condition-dependent zero-point sums). As evidence, we show: (i) the fine structure constant α^{-1} decomposes into an algebraic leading part from $\zeta(-1), \zeta(-3)$ (Bernoulli) plus transcendental corrections from $\zeta(6), \zeta(7)$ (periods) — precisely the structure predicted by the Deligne–Beilinson period conjectures; (ii) the dark energy density $\Omega_\Lambda = 16|\zeta(-1)|\zeta(2)/\pi = 2\pi/9$ combines both wings in a single observable, derivable independently from Wheeler–DeWitt quantum cosmology with temporal Dirichlet–Neumann boundaries; (iii) the Euler product truncation at $p = 2$ ($\zeta \to \zeta_{\neg 2}$), corresponding to DN boundary conditions, is a non-perturbative selection rule invisible to perturbation theory. This framework positions vacuum cosmology as the natural complement — not competitor — to Kreimer–Connes perturbative renormalization, connected by the functional equation.

1 Introduction

1.1 Kreimer–Connes: ζ values are real in perturbative QFT

The work of Kreimer [1] and Connes–Kreimer [2, 3, 4] established that renormalization in perturbative quantum field theory is not an ad hoc subtraction procedure but a precise algebraic operation: the Birkhoff decomposition of characters on a Hopf algebra of Feynman graphs. A key consequence is that the *finite* (renormalized) parts of Feynman integrals evaluate to **motivic periods** — algebraic combinations of multiple zeta values (MZVs) and polylogarithms [5, 6]:

$$\text{Renormalized Feynman integral} \in \mathbb{Q}[\zeta(3), \zeta(5), \zeta(7), \ldots, \mathrm{Li}_n(\tfrac{1}{2}), \ln 2, \pi^2]. \tag{1}$$

For example, the two-loop QED contribution to the electron anomalous magnetic moment contains $\frac{3}{4}\zeta(3)$ [8]. The three-loop involves $\zeta(5)$. Higher loops bring $\zeta(7)$ and MZVs of increasing depth.

Connes and Marcolli [7] further showed that a *cosmic Galois group* acts on these periods, giving the arithmetic content of perturbative QFT a Galois-theoretic structure analogous to the motivic Galois group of algebraic geometry.

The philosophical content of this program is profound: ζ **values at positive integers are not regularization artifacts but genuine physical invariants of quantum field theory.** They are periods of algebraic varieties (complements of graph hypersurfaces), and their appearance in physical observables (anomalous dimensions, β-functions, $g-2$) is mathematically inevitable.

1.2 The missing wing

The Kreimer–Connes framework operates within perturbation theory. It addresses the arithmetic of *loop integrals* — fluctuations *around* a background vacuum. It does not address the vacuum itself.

However, the vacuum also has arithmetic content. The zero-point energy of a quantum field in a cavity of size a is formally

$$E_0 = \frac{\hbar}{2} \sum_n \omega_n \tag{2}$$

which diverges but, via ζ-regularization, yields a finite physical quantity — the Casimir energy. The regularization involves ζ at *negative integers*:

$$\zeta(-1) = -\frac{1}{12} = -\frac{B_2}{2}, \quad \zeta(-3) = \frac{1}{120} = -\frac{B_4}{4}, \quad \ldots \tag{3}$$

These are *Bernoulli numbers* — rational, algebraic, on the opposite side of the ζ function from the periods $\zeta(3), \zeta(5), \ldots$ that Kreimer–Connes handles.

The question of this paper is:

> *If Kreimer–Connes establishes that ζ at positive integers is physically real (in perturbative QFT), is ζ at negative integers also physically real (in non-perturbative vacuum physics)?*

We argue **yes**, and show that the Riemann functional equation provides the bridge.

1.3 The functional equation as a physical duality

The Riemann functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin\!\left(\tfrac{\pi s}{2}\right) \Gamma(1-s)\, \zeta(1-s) \tag{4}$$

relates $\zeta(s)$ to $\zeta(1-s)$. At $s = 2$ and $s = -1$:

$$\zeta(2) = \frac{\pi^2}{6} \quad \longleftrightarrow \quad \zeta(-1) = -\frac{1}{12}. \tag{5}$$

The left side is a transcendental period; the right side is a rational Bernoulli value. They are "dual" under the functional equation.

We propose that this mathematical duality has a *physical* counterpart:

	Perturbative wing	**Non-perturbative wing**
ζ arguments	Positive integers	Negative integers
Values	Periods (π^{2k}, MZVs)	Bernoulli ($B_{2k}/(2k)$)
Physics	Loop integrals	Vacuum zero-point
Framework	Kreimer–Connes	Wheeler–DeWitt DN
Observables	a_e, α running	Ω_Λ, Casimir
Math nature	Transcendental	Algebraic (rational)

The functional equation (4) connects these wings, and we will show that specific physical observables — notably α^{-1} and Ω_Λ — involve *both* wings simultaneously.

2 The two wings in detail

2.1 Perturbative wing: Kreimer–Connes

We briefly review the Kreimer–Connes results relevant to our argument.

Hopf algebra of Feynman graphs. Let $\mathcal{H}$ be the commutative Hopf algebra generated by one-particle irreducible Feynman graphs, with coproduct encoding subdivergence structure [1]. A *character* $\phi : \mathcal{H} \to \mathcal{A}$ (where $\mathcal{A}$ is an algebra of Laurent series in the regularization parameter ε) encodes the evaluation of each graph.

Birkhoff decomposition. The character ϕ admits a unique factorization $\phi = \phi_-^{-1} \star \phi_+$, where ϕ_- contains the poles (counterterms) and ϕ_+ is holomorphic (renormalized values) [3]. This is the Bogoliubov–Parasiuk–Hepp–Zimmermann (BPHZ) renormalization in algebraic form.

Periods. The renormalized values $\phi_+(\Gamma)$ for primitive graphs Γ evaluate to periods of mixed Tate motives over $\mathbb{Z}$ [5]. Concretely, they are $\mathbb{Q}$-linear combinations of products of $\zeta(n)$ ($n \geq 2$ integer) and $\ln 2$. For massless propagators, only odd ζ-values $(\zeta(3), \zeta(5), \ldots)$ appear at the single-zeta level.

Physical consequence. The QED electron $g-2$, computed to five loops, contains $\zeta(3), \zeta(5)$ at 2–3 loops, and deeper MZVs at higher loops [8, 9]. These are *measured* through the electron $g-2$ experiment, making them empirically verified physical quantities.

2.2 Non-perturbative wing: vacuum zero-point

The vacuum zero-point energy (2) depends on *boundary conditions* imposed on the field modes:

- **Dirichlet–Dirichlet (DD)** or **Neumann–Neumann (NN)**: All-integer modes $n = 1, 2, 3, \ldots$. Regularized sum $\sum n^{-s}|_{s=-1} = \zeta(-1) = -1/12$.
- **Dirichlet–Neumann (DN)**: Odd-integer modes $n = 1, 3, 5, \ldots$. Regularized sum $\sum_{\text{odd}} n^{-s}|_{s=-1} = \zeta_{\neg 2}(-1) = +1/12$.

where

$$\zeta_{\neg 2}(s) \equiv (1 - 2^{-s})\zeta(s) \tag{6}$$

is the ζ function with the $p = 2$ Euler factor removed.

Key difference from Kreimer–Connes: this is *not* a loop integral. It is the sum over all modes of the *background vacuum*, without perturbative expansion. The ζ values at negative integers $(\zeta(-1), \zeta(-3), \ldots)$ emerge not from Feynman diagrams but from the *mode spectrum* of the vacuum itself.

Physical consequence. The 3D Casimir effect gives a measurable force $\propto \zeta(-3) = 1/120$ [11], confirmed experimentally [12]. Boyer's DN Casimir prediction involves $\zeta_{\neg 2}(-3) = -7/120$ [13], not yet experimentally verified.

2.3 Why Kreimer–Connes cannot see the non-perturbative wing

Perturbation theory expands around a *fixed* vacuum. The vacuum itself is the zeroth-order background. Its zero-point energy is subtracted (normal ordering) or absorbed into the cosmological constant. Kreimer–Connes, operating within perturbation theory, systematically handles all loop corrections *above* the vacuum, but the vacuum energy itself lies *outside* the Hopf algebra of Feynman graphs.

This is precisely why perturbative QFT cannot solve the cosmological constant problem: Λ is a non-perturbative vacuum property, not a loop correction.

3 Evidence: α^{-1} spans both wings

3.1 Decomposition

We have previously reported [19] the arithmetic formula

$$\alpha^{-1} \approx \underbrace{\frac{2}{|B_2|} + \frac{4}{|B_4|}}_{\text{algebraic (Bernoulli)}} + \underbrace{g \cdot 132 + 4(\zeta(6) - \zeta(7))}_{\text{transcendental (periods)}} \tag{7}$$

where $2/|B_2| = 12$, $4/|B_4| = 120$, $g \approx 0.038$, and $4(\zeta(6) - \zeta(7)) \approx 0.036$.

We now observe that this decomposition has a natural interpretation in terms of the two wings:

- **Leading (algebraic) part**: $12 + 120 = 132$. These are $1/|\zeta(-1)| + 1/|\zeta(-3)|$ — values at *negative* integers, the non-perturbative wing. They represent the "vacuum contribution" to α^{-1}.
- **Correction (transcendental) part**: $g \cdot 132$ and $\zeta(6) - \zeta(7)$. These involve periods at *positive* integers, the perturbative wing. They represent the "loop corrections" to α^{-1}.

3.2 Deligne–Beilinson structure

The Deligne period conjecture [15] and Beilinson conjectures [16] predict that critical L-values of motives factor as

$$L^*(M, n) \sim c^+(M) \cdot \Omega(M) \tag{8}$$

where c^+ is algebraic and Ω is a period.

Formula (7) has precisely this form:

$$\alpha^{-1} = \underbrace{132}_{\text{algebraic } c^+} + \underbrace{\text{transcendental corrections}}_{\text{period } \Omega} \tag{9}$$

If this identification holds, α^{-1} is a physical instance of the Beilinson conjectures, with the algebraic part from the non-perturbative wing and the period part from the perturbative wing.

4 Evidence: Ω_Λ combines both wings

4.1 The functional-equation dual form

We have shown [20] that the dark energy density parameter

$$\Omega_\Lambda = \frac{2\pi}{9} \tag{10}$$

admits the representation

$$\Omega_\Lambda = \frac{16\,|\zeta(-1)|\,\zeta(2)}{\pi} \tag{11}$$

which explicitly combines:

- $\zeta(-1) = -1/12$: non-perturbative wing (Bernoulli, algebraic)
- $\zeta(2) = \pi^2/6$: perturbative wing (period, transcendental)
- $1/\pi$: normalization (from Einstein's equation $8\pi G$)

Ω_Λ lives on the bridge between the two wings.

4.2 Wheeler–DeWitt derivation

The derivation proceeds via a non-perturbative argument [20]:

1. The Wheeler–DeWitt wave function $\Psi(a)$ satisfies temporal Dirichlet–Neumann boundary conditions (Dirichlet at Big Bang, Neumann at de Sitter asymptotic).
2. The 1D DN zero-point energy regularizes to $\zeta_{\neg 2}(-1) = +1/12$ (non-perturbative wing, with Euler $p = 2$ truncation).
3. The Cohen–Kaplan–Nelson holographic bound [10] provides $\rho_\Lambda \leq M_P^2/\ell_H^2$ (the bridge between UV and IR).
4. Saturation with coefficient $1/12$ yields $\Omega_\Lambda = 2\pi/9$.

Steps 1–2 are purely non-perturbative (no loops). Step 3 involves the holographic principle (UV/IR mixing). The final result (11) involves $\zeta(2)$ (perturbative wing) via the 8π of Einstein's equation.

4.3 Why perturbation theory cannot produce Ω_Λ

This explains why the cosmological constant problem has resisted perturbative solution for 40+ years [14]. The vacuum energy ρ_Λ is a non-perturbative quantity (ζ at negative integers) that perturbation theory (ζ at positive integers) cannot access. Kreimer–Connes, being perturbative, is structurally unable to predict Ω_Λ. The non-perturbative wing is required.

5 The $p = 2$ Euler truncation: structural selection theorems

5.1 DN boundary and $\zeta_{\neg 2}$

The Euler product of ζ is

$$\zeta(s) = \prod_{p \text{ prime}} (1 - p^{-s})^{-1}. \tag{12}$$

Removing the $p = 2$ factor gives

$$\zeta_{\neg 2}(s) = (1 - 2^{-s})\zeta(s) = \prod_{p \geq 3} (1 - p^{-s})^{-1}. \tag{13}$$

Physically, $\zeta_{\neg 2}$ arises from DN boundary conditions (odd modes only). The removal of the $p = 2$ Euler factor corresponds to the *elimination of even harmonics* by the mixed boundary. In what follows, we show that $p = 2$ is *not a choice* but a *structural necessity*, selected by multiple independent constraints.

5.2 Physicality theorem: $p = 2$ is unique

For a general single-prime removal $\zeta_{\neg p}(s) = (1 - p^{-s})\zeta(s)$, the implied dark energy parameter is

$$\Omega_\Lambda(p) = \frac{8\pi}{3}|\zeta_{\neg p}(-1)| = \frac{8\pi}{3} \cdot \frac{p-1}{12} = \frac{2\pi(p-1)}{9}. \tag{14}$$

Theorem 1 (Unique physicality of $p = 2$). *Among all primes p, the requirement $\Omega_\Lambda(p) \in [0, 1]$ (flat-universe physicality) is satisfied if and only if $p = 2$.*

Proof. $\Omega_\Lambda(p) < 1$ requires $(p - 1) < 9/(2\pi) \approx 1.432$. Since $p \geq 2$ is prime, $p - 1 \geq 1$. The constraint $1 \leq p - 1 < 9/(2\pi)$ admits only $p - 1 = 1$, i.e., $p = 2$. For $p = 3$: $\Omega_\Lambda(3) = 4\pi/9 \approx 1.396 > 1$. For $p \geq 5$: $\Omega_\Lambda(p) > 2$. □

This theorem is *independent of the observed value* $\Omega_\Lambda^{\text{obs}} = 0.685$. It depends only on (i) the Euler product structure of ζ, (ii) the CKN bound form $\Omega_\Lambda \propto |\zeta_{\neg p}(-1)|$, and (iii) the flat-universe constraint $\Omega_\Lambda < 1$.

5.3 Multi-prime removal: sign exclusion

Theorem 2 (Sign exclusion)**.** *Removing exactly two primes from the Euler product at $s = -1$ yields* negative *vacuum energy:*

$$\zeta_{\neg\{p,q\}}(-1) = (1-p)(1-q)\zeta(-1) < 0 \quad \textit{for all primes } p, q \geq 2. \tag{15}$$

Proof. $(1-p) < 0$ and $(1-q) < 0$ for primes $p, q \geq 2$, so $(1-p)(1-q) > 0$. Multiplied by $\zeta(-1) = -1/12 < 0$, the result is negative. □

Negative vacuum energy produces cosmic deceleration, contradicting observation. Therefore *exactly one* prime must be removed. Combined with Theorem **??**, this gives:

Corollary 1. *The observed positive, sub-unity dark energy density $\Omega_\Lambda \in (0, 1)$ uniquely selects the removal of the single prime $p = 2$ from the Euler product.*

5.4 Minimum energy principle

Even without the CKN normalization, $p = 2$ is selected by a variational principle:

Proposition 1 (Minimum positive vacuum energy)**.** *Among all single-prime removals, $p = 2$ gives the minimum positive vacuum energy:*

$$|\zeta_{\neg p}(-1)| = \frac{p-1}{12} \geq \frac{1}{12}, \quad \textit{with equality iff } p = 2. \tag{16}$$

The next prime $p = 3$ gives $|\zeta_{\neg 3}(-1)| = 1/6$, exactly twice as large. The hierarchy grows: for $p = 5$, four times; for $p = 7$, six times. At general negative odd integer $s = -n$, the ratio $|\zeta_{\neg 3}(-n)|/|\zeta_{\neg 2}(-n)| = (3^n - 1)/(2^n - 1) \to (3/2)^n$, growing *exponentially*. The preference for $p = 2$ is not marginal but asymptotically dominant.

5.5 Dimensional entanglement: why 3 spatial dimensions

The physicality bound generalizes to d spatial dimensions. In d dimensions, the CKN-type relation gives $\Omega_\Lambda(p, d) \propto (p-1) \cdot d^2/(2\pi \cdot 12) \cdot$ (geometric factors). The precise generalization of Theorem **??** yields:

$$p - 1 < \frac{d^2}{2\pi} \tag{17}$$

as the condition for $\Omega_\Lambda < 1$.

Observation 1 (3D–p=2 joint selection)**.** *For $p = 2$ ($p-1 = 1$), the bound (**??**) requires $d^2 > 2\pi$, i.e., $d > \sqrt{2\pi} \approx 2.507$, so $d \geq 3$. For $d = 2$: $d^2/(2\pi) = 4/(2\pi) \approx 0.637 < 1$, so* no prime works. *For $d = 1$: $1/(2\pi) \approx 0.159$, again no prime works.*

This means **3 spatial dimensions and $p = 2$ selection are not independent facts**: they are jointly constrained by a single inequality. The universe must have $d \geq 3$ for any prime truncation to produce physical dark energy, and given $d = 3$, only $p = 2$ works.

5.6 K-theoretic interpretation

The Euler factor at $p = 2$ and $s = -1$ is $(1 - 2^{-(-1)}) = 1 - 2 = -1$. This is the *unique generator* of $K_1(\mathbb{Z}) = \mathbb{Z}/2$. For any other prime p, $(1 - p)$ has $|1 - p| > 1$, exceeding the K_1 generator magnitude.

The sign flip $\zeta(-1) \to \zeta_{\neg 2}(-1)$ is thus *multiplication by the $K_1(\mathbb{Z})$ generator* (-1) — the unique non-trivial element of the group of units modulo torsion. This connects the vacuum sign flip to the spin structure of spacetime (Spin $\to SO$ is a $\mathbb{Z}/2$ double cover), providing an algebraic-topological origin for the $p = 2$ selection.

5.7 Invisibility to perturbation theory

In perturbative QFT, all modes contribute to loop integrals. There is no mechanism to select odd modes only. The Euler product truncation at $p = 2$ is a *boundary-condition effect* on the vacuum, not a loop-level phenomenon.

Proposition 2 (Non-perturbative character of $\zeta_{\neg 2}$)**.** *The transition $\zeta \to \zeta_{\neg 2}$ (equivalently, DD $\to$ DN boundary) is invisible to perturbation theory. It is a non-perturbative selection rule operating at the vacuum level.*

This is why Kreimer–Connes never encounters $\zeta_{\neg 2}$. The perturbative wing uses the *full* ζ (all primes). The non-perturbative wing uses $\zeta_{\neg 2}$ (odd primes only) when DN boundaries are present.

5.8 Summary: five independent arguments for $p = 2$

#	Argument	Status
1	Physicality ($\Omega_\Lambda < 1$)	Theorem **??**
2	Sign (single removal $\to$ positive)	Theorem **??**
3	Minimum energy (|min at p=2|)	Proposition
4	Dimensional ($d \geq 3 \Leftrightarrow p$=2 works)	Observation
5	$K_1(\mathbb{Z}) = \mathbb{Z}/2$ generator	Algebraic

These five arguments are logically independent and use different mathematical structures (analysis, sign combinatorics, optimization, dimension theory, algebraic K-theory). Their convergence on $p = 2$ constitutes *structural* evidence, not numerical coincidence.

6 Unified picture

6.1 The zeta building

We propose the metaphor of a "zeta building" with two wings:

$$\underbrace{\zeta(-5), \zeta(-3), \zeta(-1)}_{\text{Non-perturbative wing}} \quad [\text{functional equation}]\zeta(s) \leftrightarrow \zeta(1-s) \quad \underbrace{\zeta(2), \zeta(3), \zeta(5), \zeta(7)}_{\text{Perturbative wing}} \tag{18}$$

- **Left (non-perturbative) wing**: Bernoulli numbers, vacuum zero-point, Casimir effect, Ω_Λ, boundary-condition dependence. Algebraic values. Accessible via Wheeler–DeWitt quantum cosmology.
- **Right (perturbative) wing**: Periods, MZVs, loop integrals, anomalous dimensions, $g-2$. Transcendental values. Accessible via Kreimer–Connes Hopf algebra.
- **Connecting hallway**: The Riemann functional equation. Physical observables like α^{-1} and Ω_Λ span both wings.

6.2 Physical observables and their wing structure

Observable	Left wing	Right wing	Bridge
a_e (electron $g-2$)	—	$\zeta(3), \zeta(5)$	KC only
Casimir (DD)	$\zeta(-3)$	—	Left only
Casimir (DN)	$\zeta_{\neg 2}(-3)$	—	Left only
Ω_Λ	$\zeta_{\neg 2}(-1)$	$\zeta(2)$ via $8\pi G$	Both
α^{-1}	$1/\lvert\zeta(-1)\rvert, 1/\lvert\zeta(-3)\rvert$	$\zeta(6), \zeta(7)$	Both

Some observables live purely in one wing; others span both. The spanning observables (α^{-1}, Ω_Λ) are the most physically fundamental, as they require the full ζ building for their description.

7 Relation to existing programs

7.1 Kreimer–Connes: the perturbative wing is established

The right wing is mathematically rigorous and experimentally verified (electron $g-2$ to $\sim 10^{-13}$ precision). Our proposal does not modify or challenge Kreimer–Connes. It *extends* the same philosophy ("ζ values are physically real") to the non-perturbative side.

7.2 Casimir physics: the left wing is partially established

The Casimir effect for DD boundaries is experimentally confirmed [12]. The $\zeta(-3) = 1/120$ coefficient appears in the measured force. Boyer's DN prediction ($\zeta_{\neg 2}(-3) = -7/120$) is not yet experimentally tested.

Our proposal extends this to cosmological scales via the Wheeler–DeWitt temporal DN argument.

7.3 Deligne–Beilinson: the bridge is conjectured

The Deligne and Beilinson conjectures [15, 16] predict that L-values factor into algebraic and transcendental (period) parts. Our observation that α^{-1} and Ω_Λ have precisely this structure is a candidate physical realization.

7.4 Moonshine: a parallel

Monstrous Moonshine [17] demonstrated that modular forms (j-function) encode genuine physics (Monster VOA, 2D CFT). Mathieu Moonshine [18] extended this to K3 surfaces. Our proposal that ζ special values encode 4D physical constants is a parallel "ζ-moonshine" — not via a sporadic group, but via the functional equation duality of the ζ function itself.

8 Predictions

The unified framework makes specific testable predictions:

1. $\Omega_\Lambda = 2\pi/9$: Testable at 0.5% by Euclid/DESI/LSST (2025–2028). This is a purely non-perturbative-wing prediction, inaccessible to perturbation theory.

2. **DN Casimir (Boyer)**: The 3D DN Casimir coefficient $\zeta_{\neg 2}(-3) = -7/120$ (repulsive) can be tested in laboratory experiments with mixed-boundary acoustic or electromagnetic resonators.

3. α^{-1} **decomposition**: The split of α^{-1} into algebraic (132) and transcendental corrections can be compared against high-precision QED calculations. If the split is exact, higher-loop terms should organize into period-valued corrections to the Bernoulli leading part.

4. **Muon and tau** $g-2$: The identification $\Delta a_\mu \approx 252 = 6/|B_6|$ (non-perturbative wing) and the Ramanujan discriminant $\tau(3) = 252$ suggest that generational anomalies may bridge both wings. Prediction: $\Delta a_\tau \propto \tau(5) = 4830$.

9 Limitations

1. The "two-wing" picture is a *proposal*, not a theorem. The functional equation is a mathematical identity; its physical interpretation as a duality between perturbative and non-perturbative physics is conjectural.

2. The Wheeler–DeWitt derivation of Ω_Λ assumes specific boundary conditions (temporal DN) and CKN bound saturation. These assumptions are physically motivated but not derived from first principles.

3. The decomposition of α^{-1} into algebraic + period parts is observed post hoc. A rigorous derivation via Kreimer–Connes Hopf algebra is future work.

4. The Euler $p = 2$ truncation (DN boundary) is presented as a non-perturbative selection rule, but its fundamental origin (why $p = 2$? why DN?) remains open. Candidates include spin structure ($K_1(\mathbb{Z}) = \mathbb{Z}/2$) and time-reversal asymmetry.

5. This framework does not address fermion masses, Yukawa couplings, or CKM mixing — these may require additional structure beyond the ζ building.

10 Discussion

The central thesis of this paper is that the arithmetic content of quantum physics is organized by the Riemann ζ function, with perturbative physics on one side of the functional equation and non-perturbative vacuum physics on the other.

Kreimer–Connes brilliantly established the perturbative side. They showed that ζ values at positive integers are not regularization artifacts but genuine motivic periods with Galois-theoretic structure. Our contribution is to extend this philosophy to the other side: ζ values at negative integers (Bernoulli numbers) appearing in vacuum energy regularization are not computational conveniences but genuine physical data about the boundary-condition structure of the vacuum.

If this picture is correct, the cosmological constant problem is not a failure of quantum field theory per se, but a consequence of *looking only at the perturbative wing*. The full ζ building, spanning both wings via the functional equation, naturally produces the observed small positive Λ from the non-perturbative wing ($\zeta_{\neg 2}(-1) = +1/12$) combined with the holographic bridge (CKN bound).

The beauty of this picture lies in its economy: no new fields, no extra dimensions, no multiverse. Only the recognition that the ζ function's functional equation is not merely a mathematical symmetry but a *physical duality* between the perturbative and non-perturbative sectors of quantum physics.

11 Conclusion

We have proposed that the physical content of the Riemann ζ function comprises two complementary wings:

- The **perturbative wing** (ζ at positive integers: periods, MZVs, loop integrals), established by Kreimer–Connes.
- The **non-perturbative wing** (ζ at negative integers: Bernoulli numbers, vacuum zero-point, boundary conditions), proposed by this work via Wheeler–DeWitt quantum cosmology.

The Riemann functional equation connects these wings, and specific physical observables (α^{-1}, Ω_Λ) span both. The framework provides:

1. A parameter-free prediction $\Omega_\Lambda = 2\pi/9$ (2% agreement with Planck 2018).
2. A Deligne–Beilinson-type decomposition of α^{-1} into algebraic and period parts.
3. An explanation of why the cosmological constant problem resists perturbative solution (it lives in the non-perturbative wing).
4. A non-perturbative selection rule ($\zeta \to \zeta_{\neg 2}$, DN boundary) invisible to loop expansion.

The proposal positions vacuum cosmology as a natural complement to Kreimer–Connes perturbative renormalization. Together, they may constitute the two halves of a complete arithmetic description of quantum physics — united by the functional equation of the Riemann ζ function.

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